

Cybersecurity

Week: 1

Topic: Fundamentals of Cybersecurity and Online Threats

1. Overview

Cybersecurity refers to the practice of protecting systems, networks, and programs from digital attacks. These attacks aim to steal sensitive information, disrupt operations, or damage systems. With the increasing reliance on technology, individuals, businesses, and governments are more exposed to cyber threats.

Understanding the basics of cybersecurity is essential for anyone using digital platforms. It is not only about technology but also about human behavior, processes, and awareness.

2. Key Concepts

A. Common Types of Cyber Threats :

Malware: Malicious software that can damage or disrupt systems. Examples include viruses, ransomware, spyware, and trojans.

Phishing: Attempts to trick users into providing sensitive data such as passwords or credit card numbers through fake emails, messages, or websites.

Denial-of-Service (DoS) Attacks: Attackers overwhelm a system or network with traffic to make services unavailable.

Man-in-the-Middle (MITM) Attacks: Attackers intercept communication between two parties to steal, modify, or manipulate data.

Social Engineering: Exploiting human psychology to gain unauthorized access, such as pretending to be IT support to get passwords.

B. Human Factor

Many cybersecurity incidents occur because of human error rather than purely technical failures. Weak passwords, sharing sensitive information, or falling for phishing scams are common causes of breaches. Promoting awareness and training users are essential components of a strong cybersecurity strategy.

C. Cybersecurity Principles (CIA Triad)

Confidentiality: Ensuring that information is accessible only to authorized users.

Example: Using encryption to protect sensitive customer data.

Integrity: Maintaining the accuracy and consistency of data. Example: Preventing unauthorized changes to financial records.

Availability: Ensuring systems and information are accessible when needed. Example: Regular backups and disaster recovery plans to prevent downtime.

D. Cyber Hygiene Best Practices

Use strong, unique passwords and update them regularly.

Enable multi-factor authentication (MFA) wherever possible.

Keep operating systems, apps, and antivirus software updated.

Avoid clicking on suspicious links or downloading unverified files.

Educate all users in an organization about phishing and other scams.